Phase 0 — Freeze the Current App (1 day)

Before adding AI, make sure the existing foundation is stable.

- ✅ Expense CRUD
- ✅ Budget CRUD
- ✅ Charts
- ✅ Authentication
- ✅ Expense Categorization ML
- ✅ Dashboard

No new features until this is clean.

Phase 1 — Analytics Engine ⭐ (Highest Priority)

This is the heart of SIA.

Don't write a single line of LLM code yet.

Create a backend module:

```
backend/

analytics/

    spendingAnalyzer.js

    budgetAnalyzer.js

    categoryAnalyzer.js

    trendAnalyzer.js

    habitAnalyzer.js

    healthAnalyzer.js

    reportGenerator.js
```

Each analyzer has one responsibility.

Example:

```
        Expenses
           |
           ▼
   Spending Analyzer
           |
           {
     totalExpense,
   averageDailySpend,
   averageWeeklySpend
           }
```

Another:

```
        Expenses
           |
           ▼
   Category Analyzer
           |
           {
      topCategory,
     leastCategory,
  categoryDistribution
           }
```

Finally:

```
   Report Generator
           ↓
   Merge everything
           ↓
   Financial Report
```

This report becomes the single source of truth.

Phase 2 — Define the Financial Report

This is probably the most important file in the project.

Something like:

```
{
  "summary": {},

  "budgets": {},

  "categories": {},
```

"trends": {},

"habits": {},

"financialHealth": {},

"forecast": {},

"risk": {}
}

Everything in BALENISA should use this report.

- Dashboard
- Insights
- Notifications
- SIA

Everyone reads from the same report.

Phase 3 — Forecasting ML

Now build another ML service.

Input:

Past expenses

Output:

Expected month-end spending

This becomes another section in the report.

Phase 4 — SIA Backend

Now create

backend/

sia/

Inside

intentClassifier.js
contextBuilder.js
promptBuilder.js
llmService.js
responseFormatter.js

Notice

No frontend yet.

Just backend.

Phase 5 — Intent Detection

User asks

How much did I spend on food?

Intent

CATEGORY_QUERY

User asks

Predict this month's spending

Intent

FORECAST_QUERY

User asks

Why are my expenses increasing?

Intent

ANALYSIS_QUERY

Don't involve the LLM if the backend can answer directly.

Phase 6 — Context Builder

This is where your backend becomes intelligent.

Example

Question

Why am I overspending?

Context Builder collects

- Budget usage
- Trend analysis
- Category growth
- Forecast

Only these.

Not the whole database.

Phase 7 — LLM

Now connect

- OpenAI
- Gemini
- Claude
- Local model

Prompt

System Prompt
+
Financial Report
+
Relevant Context
+
User Question

Output

Natural answer.

Phase 8 — Frontend Chat

Only now.

Floating SIA Button
↓
Chat Window
↓
Backend
↓
SIA

Phase 9 — Voice

Mic
↓
Speech to Text
↓
SIA
↓
Text
↓
Speech

Phase 10 — Smart Notifications

Examples

Budget Risk
Forecast
Weekend Overspending
Recurring Merchant Alerts

Final Architecture

React

Express API

Expense Service
Budget Service
Analytics Engine
Forecast ML
Notification Engine

Financial Report

Intent Detection

Context Builder

SIA

LLM Response